

Assignment 4

Portfolio Replication

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Assignment 4: 500 Portfolio Replication

Benchmark: ^GSPC daily log returns; Train: Jan 2020 to Dec 2024; Validation: Jan 2025 to Jun 2025; Holdout: Jul 2025 to Dec 2025

1. Problem Setup, Data Pipeline, and Model Design

The task is to replicate the S&P 500 daily return profile with a sparse long-only portfolio under the cardinality constraint $k \leq 50$. After validating the raw OHLCV files, we retained 578 readable stock histories, excluded 6 issue files (5 empty CSVs and one non-price cache file), removed 24 late entrants / early exits, and dropped one additional ticker (CVG) because it still contained 25 missing training prices. The final model-ready matrix covers 2020-01-03 to 2025-12-31 and contains 553 stocks plus the benchmark. Validation and holdout price holes were forward-filled before return construction, while the training block was kept hole-free by design.

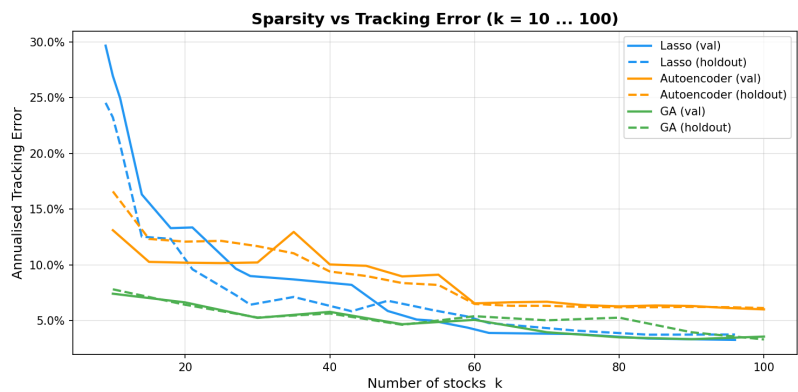
The cardinality-constrained tracking problem is an **MIQP** and is therefore **NP-hard**. To keep the comparison fair, all methods use the same two-stage pipeline: (i) method-specific stock selection and (ii) a shared long-only quadratic program that minimizes the variance of active returns subject to $\sum_i w_i = 1$ and $w_i \geq 0$. We report annualized **Tracking Error (TE)** and **Information Ratio (IR)**, where **TE** should be low and **IR** should stay close to zero for a true index tracker.

Method	Selection Stage	Implementation Note
Lasso	Non-negative Lasso regression on training benchmark returns; alpha swept on a log-grid to produce sparse portfolios of different k .	Fast, interpretable, and directly supervised by the benchmark, but sensitive to regime shocks and concentration.
Autoencoder	Symmetric autoencoder ($553 \rightarrow 128 \rightarrow 32 \rightarrow 128 \rightarrow 553$); stocks ranked by communality (reconstruction R^2) and top- k retained.	Captures shared latent structure, but unsupervised ranking can over-concentrate in sectors with strong common factor loadings.
Genetic Algorithm	Population search over k -stock subsets; fitness equals negative training TE , with crossover, mutation, and shared QP reweighting.	Computationally heavier, but best at directly optimizing sparse subset quality under the replication objective.

2. k = 50 Summary Results

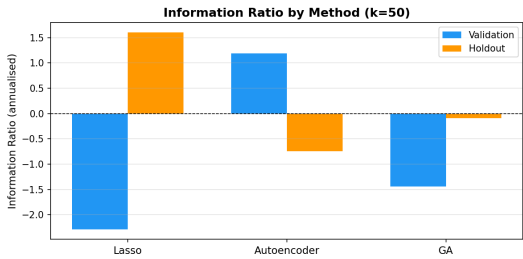
Method	TE Val	IR Val	TE Holdout	IR Holdout	Interpretation
Lasso	5.86%	-2.29	6.78%	+1.60	Moderate tracking error, but a large IR sign flip reveals unstable active tilts under the April 2025 tariff shock and subsequent rebound.
Autoencoder	8.96%	+1.19	8.37%	-0.74	Highest TE at $k = 50$; the unsupervised ranking produces persistent sector concentration , so the portfolio behaves less like a neutral tracker.
GA	4.68%	-1.44	4.63%	-0.10	Best overall . It delivers the lowest out-of-sample TE on both splits and the holdout IR closest to the ideal target of zero.

3. Sparsity vs Tracking Error and Information Ratio



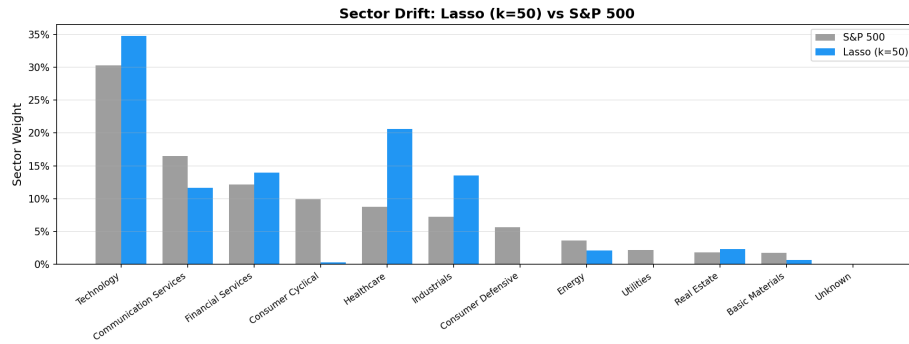
Out-of-sample **TE** falls steeply as k rises from 10 to roughly 50, then flattens. **GA** dominates the sparse end of the trade-off curve.

The tracking error vs sparsity curve shows the expected **diminishing-return pattern**: all methods improve rapidly from $k = 10$ to around $k = 50$, after which **TE** reductions become incremental. **Lasso** is especially brittle at small k (**TE** above 25% at $k \approx 10$), while **GA** already remains below 8% at $k = 10$, indicating better subset quality under severe sparsity. The **IR** plot highlights that low **TE** alone is not sufficient: **Lasso** and **Autoencoder** both exhibit unstable directional bets across validation and holdout, whereas **GA**'s holdout **IR** of -0.10 is the clearest evidence of **genuine replication** rather than accidental active management.

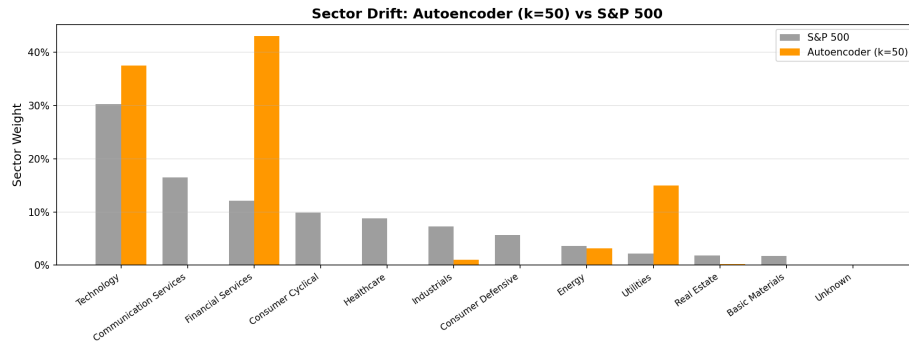


IR quantifies active drift. For a tracker, values near 0 are preferred; **GA** is closest on holdout.

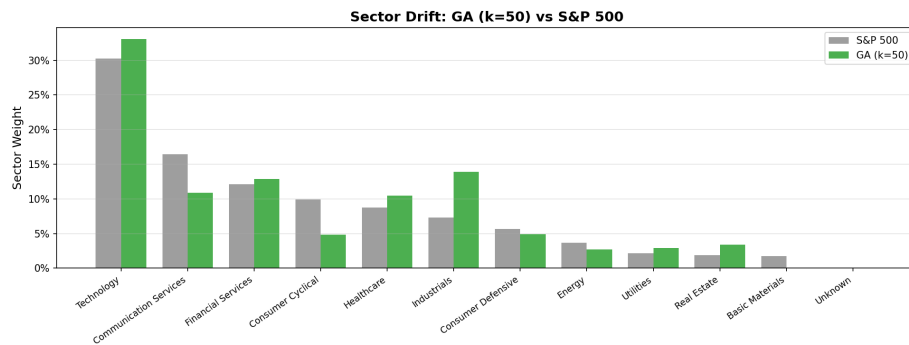
4. Sector Drift at $k = 50$



Lasso: moderate drift, with visible Healthcare overweight and weaker Communication Services exposure.



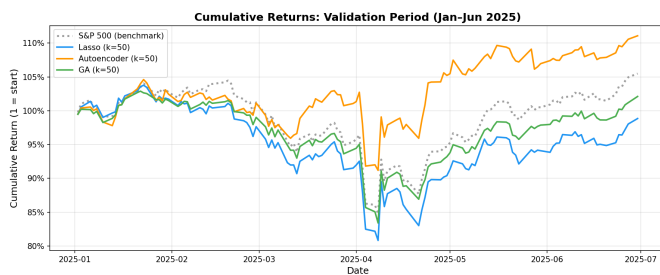
Autoencoder: the strongest sector concentration, especially in Financial Services and Utilities.



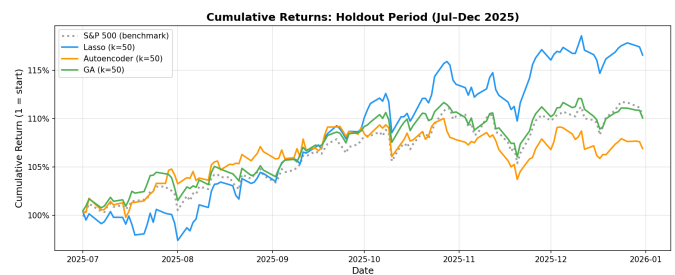
GA: the closest overall match to market-cap weighted S&P 500 sector shares.

Sector drift explains a large part of the ranking at $k = 50$. **GA** passes the **benchmark-alignment check** best, with only a mild Technology overweight and no severe sector omission. **Lasso** remains serviceable but carries visible Healthcare and Communication Services tilts. **Autoencoder** is the weakest replicator because its **communality ranking** heavily clusters in Financial Services and Utilities, producing a portfolio that behaves more like an **active thematic basket** than a neutral index proxy.

5. Cumulative Returns on Validation and Holdout



Validation: the **April 2025 tariff shock** separates robust trackers from unstable active tilts.



Holdout: all methods tighten, but **GA** remains the cleanest path around the benchmark.

The **cumulative overlays** reinforce the metric-based story. During validation, **Lasso** experiences the sharpest drawdown under the **April 2025 shock**, while **Autoencoder** overshoots the rebound and reveals a more directional active bet. **GA** stays closest to the benchmark throughout the stress period. In holdout, all three methods improve, but **GA** remains the most stable and benchmark-like, whereas **Autoencoder** drifts lower toward year-end.

6. Conclusion

Across all methods, the **Genetic Algorithm** turned out to be a good replication method. It achieves the lowest validation and holdout **TE** at the target of $k = 50$, keeps holdout **IR** closest to zero, and exhibits the cleanest **sector alignment** with the benchmark. **Lasso** remains attractive for speed and interpretability but is vulnerable to regime-specific tilts, while the **Autoencoder**'s unsupervised selection objective produces excessive **sector concentration**. The key empirical takeaway is that, for **sparse index replication**, directly optimizing subset quality with a search-based method and then reweighting through a common long-only **QP** yields a **stable out-of-sample tracker**.
